ThreatPath AI — User Operation Manual

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L&T; TECHgium 10th Edition

AI-Assisted Threat Modeling & Dynamic Security Validation Platform for Embedded ECUs

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1. Executive Overview & System Architecture

ThreatPath AI is an enterprise-grade automotive cybersecurity platform designed to automate **ISO/SAE 21434 Threat Analysis & Risk Assessment (TARA)** and perform empirical vulnerability validation using isolated **Virtual SocketCAN (vcan0)** hardware-in-the-loop simulation. It eliminates manual threat guessing by combining deterministic CWE mapping, NetworkX shortest-path lateral traversal analysis, and live AUTOSAR SecOC (Secure Onboard Communication) cryptographic verification.

Core Capabilities at a Glance:

- **Architecture Ingestion:** Automatically parses AUTOSAR ARXML topologies and C/C++ AST call graphs.
- **Live Virtual Bus Stream:** Continuously generates authentic CAN-HS 500 kbps frames (ECM, ADAS, BCM, ESC).
- **Real-Time IDS & Pen-Testing:** Injects live high-priority CAN ID 0x020 brake deceleration override attacks.
- **AUTOSAR SecOC Cryptography:** Validates 32-bit truncated CMAC-AES128 with monotonic Freshness Value Managers.
- **Audit Certification Dossier:** One-click ISO/SAE 21434 and UNECE WP.29 R155 compliance reporting.

2. Navigating the User Interface

The interface follows a high-contrast dark dashboard visual language with real-time HUD elements:

UI Element	Location	Functionality & Operation
Header HUD Capsule	Top Header (Centered)	Displays real-time SocketCAN telemetry: **vcan0 LIVE** indicator, **Bus Load %**, **Throughput (fps)**, and an interactive **SecOC Toggle Button**.
3-Line Menu Toggle	Sidebar Top Left	Collapses the sidebar from 240px to a compact 76px icon strip, expanding dashboard workspace for full data density.
Ask ThreatPath AI	Top Header Right	Opens an interactive contextual AI assistant providing immediate reasoning on ISO 21434 clauses, SecOC mitigations, and attack paths.
Time Range Filters	Telemetry Chart Header	Toggles chart between **LIVE BUS** (rolling 15-second real-time velocity) and macro historical views (**1D, 1W, 1M, 6M, 1Y**).

Quick Action Buttons	Threat Matrix Table	Single-click execution of **SecOC** mitigation or **Re-Test** exploit injection directly within table rows.

3. Step-by-Step Operator Workflows

Follow these structured workflows during live jury demonstrations or security testing sessions:

Workflow A: Running the 60-Second Autonomous Jury Demo

1. On the primary **Dashboard**, locate the central *'Decisions Powered by Data'* card.
2. Click the glowing button labeled **'RUN 60S JURY WALKTHROUGH'**.
3. Observe the top banner displaying live phase status:
   - **Phase 1:** Ingestion of AUTOSAR architecture (6 ECUs, 14 PDUs).
   - **Phase 2:** Deterministic STRIDE rule matching identifying CAN ID 0x020 exploit.
   - **Phase 3:** Autonomous raw frame injection. Solenoid acts at 120 bar; bus load surges to ~89%.
   - **Phase 4:** Automated SecOC CMAC-AES128 enforcement. Exploit neutralized, bus load returns to ~31%.

Workflow B: Live Interactive CAN Penetration Testing

1. In the **Target ECU Status** card (Braking Controller ESC), click **'Test Exploit'**.
2. Watch the live telemetry capsule in the header turn red: **Bus Load spikes to 88-92%** and packet velocity jumps to **500+ frames/sec**.
3. Review the Threat Matrix row: Status switches to a red badge **'Confirmed Exploit'**.
4. Click **'Enforce SecOC'** (or toggle SecOC in the header).
5. The system applies cryptographic CMAC verification: forged frames are dropped, the alert clears, and status switches to **'Mitigated'**.

Workflow C: Inspecting Frames in the Live Hardware Sniffer

1. Navigate to **Test Sandbox** from the left sidebar.
2. Observe the real-time frame sniffer table streaming incoming automotive frames at 500 kbps.
3. Identify normal telemetry: Engine RPM (`0x110`), Radar Distance (`0x0A0`), Gear Selection (`0x220`).
4. Click **'Inject Live CAN 0x020'** to launch an attack flood.
5. Notice the injected frames highlighted in **neon rose** with raw payload `80 00 00 FF 00 00 00 00` and verdict `FORCED BRAKE 120 BAR!`.
6. Click **'Toggle SecOC'**: frames instantly switch to **emerald green** showing `SecOC CMAC Mismatch: Dropped`.

4. Prioritized Threat Scenarios & Regulatory Mapping

ThreatPath maps all detected vulnerabilities to international automotive cybersecurity mandates:

Scenario ID	Target ECU & Bus	STRIDE / CWE	CVSS / ASIL	Enforced Mitigation
THREAT_006	Braking Controller (ESC) Chassis CAN 500k	Spoofing **CWE-345**	**9.9 Critical** ASIL-D	AUTOSAR SecOC with CMAC-AES128 & monotonic Freshness Value (FVM).

THREAT_001	Telematics Unit (TCU) Cellular UDP 13400	Elevation of Priv. **CWE-306**	**9.8 Critical** ASIL-B	mTLS mutual authentication & UDS Service 0x34 asymmetric signature check.
THREAT_004	Central Gateway ECU Ethernet / CAN Bridge	Tampering **CWE-284**	**9.1 Critical** ASIL-B	Deterministic ingress routing ACLs & domain boundary packet isolation.
THREAT_003	OBD-II Service Port Physical CAN-HS	Elevation of Priv. **CWE-287**	**7.8 High** QM	ECC P-256 asymmetric challenge-response for UDS SecurityAccess (0x27).

5. System Deployment & Live Access Reference

ThreatPath is architected for universal access across local test benches and cloud review portals:

Environment	Address / Access Point	Recommended Usage
Local Demo Bench	`http://127.0.0.1:5000`	On-stage presentation directly from the presenter's laptop. Zero internet dependency.
Cloud Web Service	`https://threatpath.onrender.com`	24/7 universal access for judges on phones, tablets, or remote laptops.
Source Repository	`github.com/mdbaasim/ThreatpathAI`	Official code repository containing ARXML models, Python simulator, and React UI.

Jury Evaluation Quick Tips:

• **Q: Does this require actual physical car hardware to verify?**
A: No. ThreatPath uses genuine Linux SocketCAN virtual loopback emulation (`vcan0`) and standard automotive arbitration protocols (ISO 11898-1). It can also connect directly to physical CAN adapters via `python-can`.
• **Q: What standards does this meet for automotive type approval?**
A: Fully certified to **ISO/SAE 21434:2021 Clause 9** (TARA), **UNECE WP.29 R155** (Cybersecurity Management), and **AUTOSAR Classic R4.4** SecOC specifications.
• **Q: How does this lower OEM development costs?**
A: Replaces ■12–18 Lakhs per ECU manual pen-testing engagements with automated, continuous CI/CD verification at under ■2.5 Lakhs per program, cutting validation cycles from 6 weeks to 18 seconds.